

Optical Satellite Data

Vegetation Change and Phenology with Sentinel-2

From spectral indices to seasonal metrics and defensible change analysis

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From reflectance to vegetation change

A simple logic for the whole block

1 Observe

Sentinel-2 Level-2A
surface reflectance
+ pixel-level quality control

2 Describe

Vegetation indices
+ seasonal summaries
+ phenology metrics

3 Compare

Within season
Year to year
2–5 year intervals
full time series

4 Interpret

Greening / browning
moisture / stress
timing / disturbance
with uncertainty

Core rule

Do not start with “Which index should I calculate?” Start with “Which vegetation process or change do I need to detect?”

What Sentinel-2 sees in Adventdalen

A tundra pixel integrates vegetation with moisture, exposed substrate, drainage and snow history.



Landscape controls on the spectral signal

- Vegetation cover**
Sparse tundra and exposed mineral ground alter the red–NIR contrast.
- Moisture & drainage**
Wet surfaces and drainage networks change NIR–SWIR reflectance.
- Snow persistence**
Snowmelt timing can shift the apparent start of the vegetation season.

Remote-sensing implication

NDVI
green vegetation signal

NDRE
red-edge response

NDMI
moisture-sensitive context

Interpret change in the index together with the landscape setting — not as vegetation change by default.

Sentinel-2 bands for vegetation analysis

Always name the spectral range first — then the Sentinel-2 band

B2 Blue 490 nm · 10 m	B3 Green 560 nm · 10 m	B4 Red 665 nm · 10 m	B5 Red edge 1 705 nm · 20 m	B6 Red edge 2 740 nm · 20 m	B7 Red edge 3 783 nm · 20 m	B8 Near-infrared 842 nm · 10 m	B8A Narrow near-infrared 865 nm · 20 m	B11 Short-wave infrared 1 1610 nm · 20 m	B12 Short-wave infrared 2 2190 nm · 20 m
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Visible · Blue B2 · Green B3 · Red B4

Pigment absorption and visible colour. Red B4 supplies the chlorophyll-absorption term in NDVI, EVI and SAVI; Blue B2 supports the three-band EVI correction.

Red edge · B5 · B6 · B7

Samples the steep spectral transition near 700–780 nm. Useful when pigment concentration and canopy condition are part of the question.

Near-infrared · B8 · B8A

High reflectance from leaf and canopy scattering. Near-infrared B8 gives 10 m NDVI; Narrow NIR B8A pairs naturally with the 20 m red-edge and SWIR bands.

Short-wave infrared · B11 · B12

Sensitive to water content and dry material. Adds moisture and disturbance information that red–NIR greenness indices do not contain.

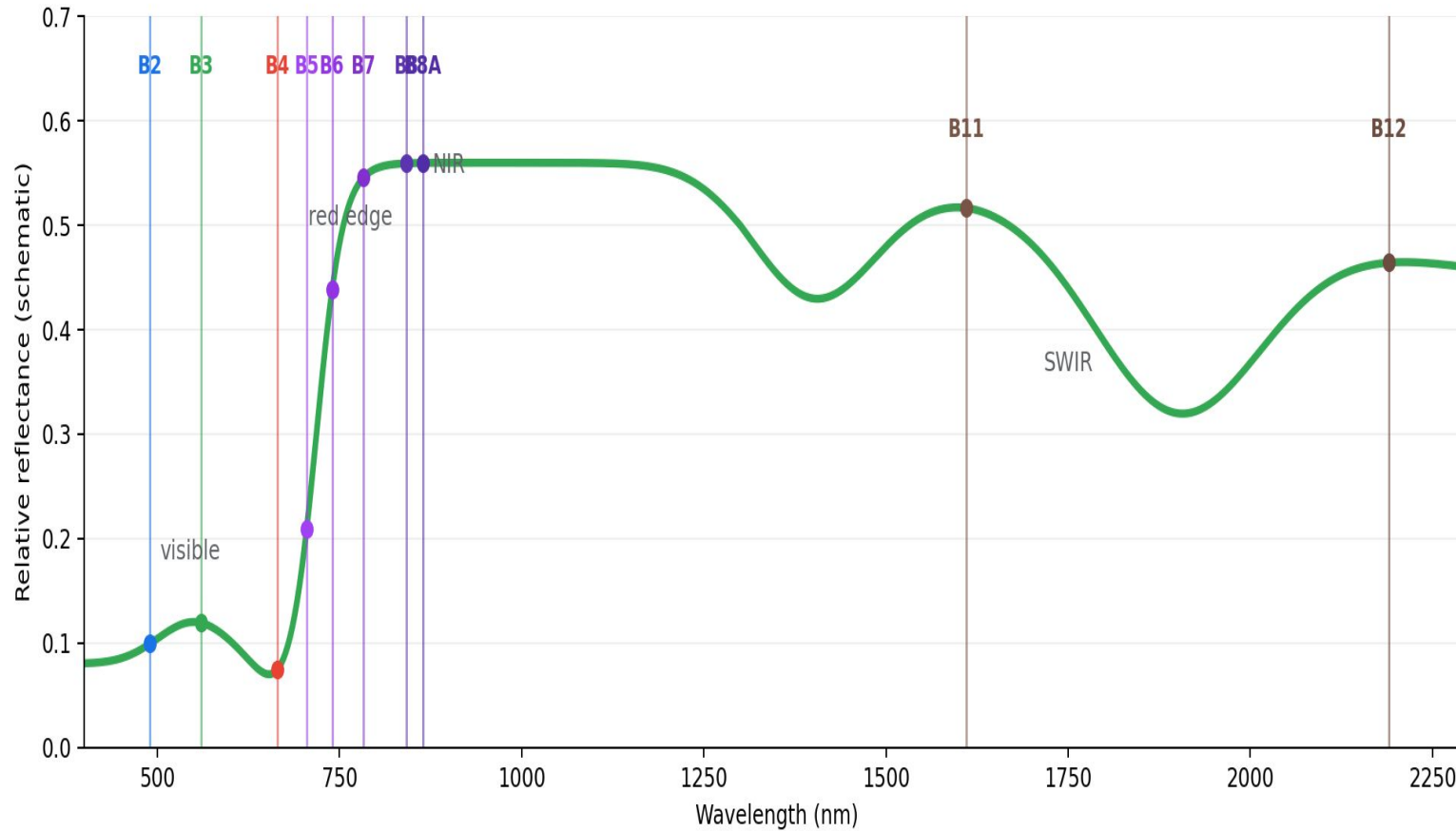
Resolution rule

An index inherits the information content of its input bands.

NDVI using Near-infrared B8 + Red B4 is a native 10 m product.
 NDRE using Narrow NIR B8A + Red edge 1 B5 is based on 20 m measurements.
 NDMI using Narrow NIR B8A + SWIR1 B11 is also based on 20 m measurements.

Why vegetation indices work

Indices emphasise spectral contrasts linked to pigments, canopy structure and water



Red absorption

Chlorophyll strongly reduces reflectance in the Red spectral range — Sentinel-2 Red B4.

Near-infrared scattering

Leaf internal structure and canopy architecture raise reflectance in the Near-infrared — B8 / B8A.

Red-edge transition

The steep Red → NIR transition sampled by B5–B7 responds to pigment concentration and canopy condition.

SWIR absorption

Short-wave infrared B11 / B12 responds strongly to water content and dry material.

An index is a proxy — it does not isolate one biological variable perfectly.

NDVI — Normalised Difference Vegetation Index

Why Red B4 and Near-infrared B8 form the standard Sentinel-2 vegetation baseline

$$\text{NDVI} = (\text{Near-infrared B8} - \text{Red B4}) / (\text{Near-infrared B8} + \text{Red B4})$$

Near-infrared · B8 · 842 nm

Red · B4 · 665 nm

What it maps

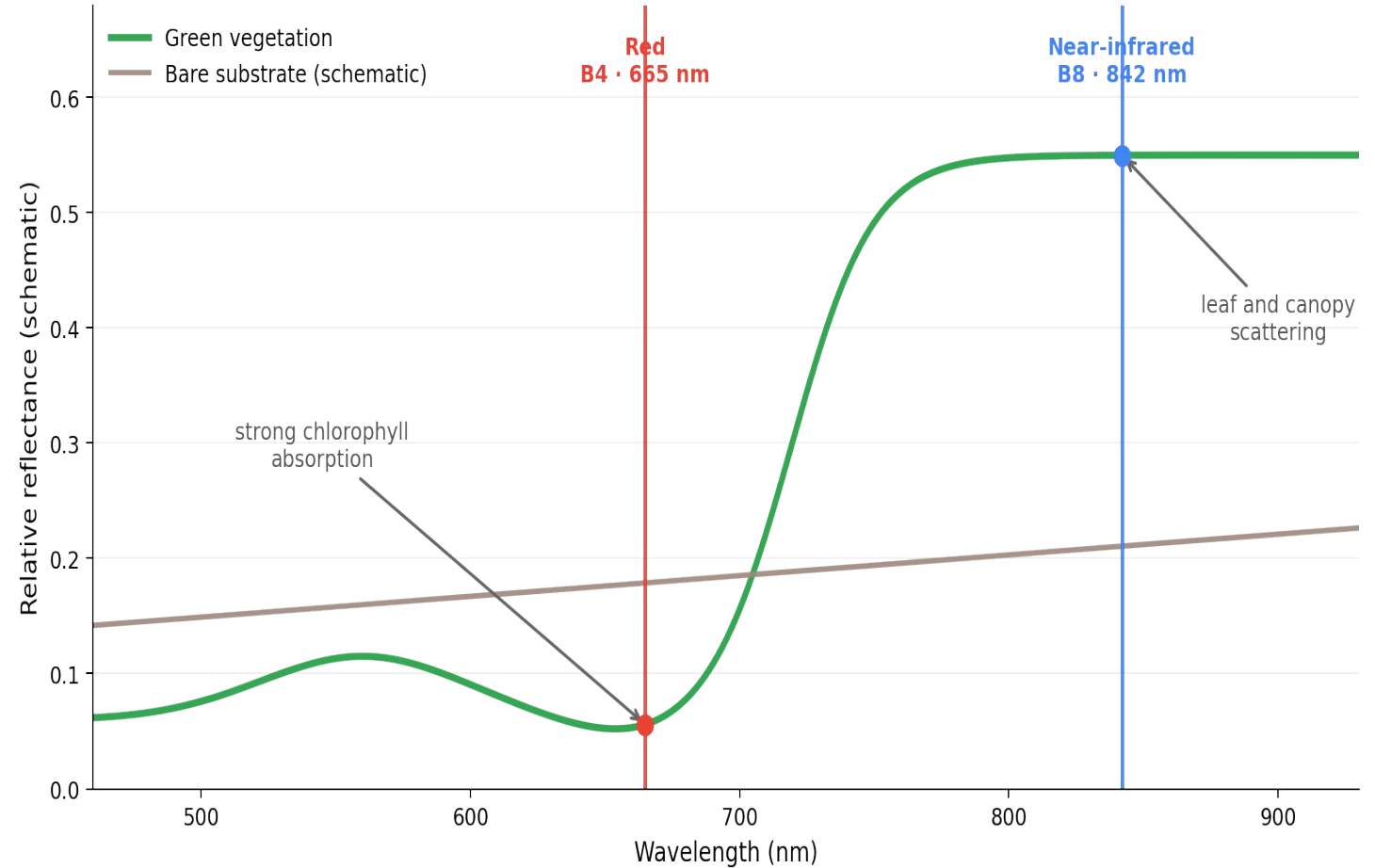
Relative green-vegetation signal, seasonal canopy development and spatial greenness differences.

Why these spectral ranges

Red B4 captures strong chlorophyll absorption; Near-infrared B8 captures leaf and canopy scattering. NDVI measures the contrast.

Best Arctic role

Baseline for tundra greenness, greening/browning and phenology — after cloud and snow screening.



Limitations: exposed substrate in sparse tundra · snow · mixed pixels · reduced sensitivity in very dense canopies

Conceptual spectrum, not measured reflectance. Band centres: Copernicus Sentinel-2 documentation.

NDRE — Normalised Difference Red Edge Index

A complementary Sentinel-2 index for pigment and canopy-condition differences

$$\text{NDRE} = (\text{Narrow NIR B8A} - \text{Red edge 1 B5}) / (\text{Narrow NIR B8A} + \text{Red edge 1 B5})$$

Narrow near-infrared · B8A · 865 nm

Red edge 1 · B5 · 705 nm

What it helps map

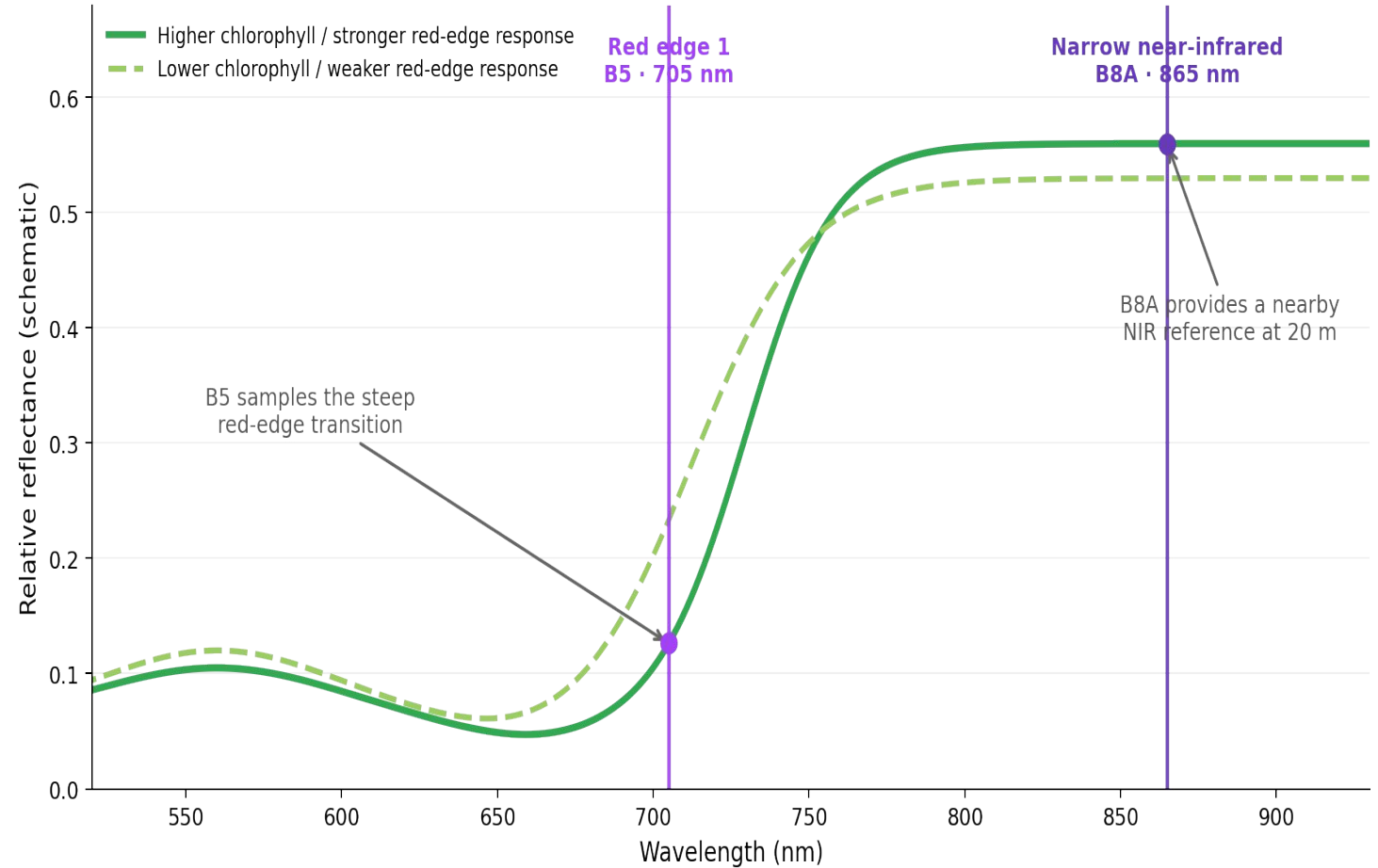
Relative red-edge response linked to pigment concentration and canopy condition; useful for vegetation-condition and community differences.

Why these spectral ranges

Red edge 1 B5 sits on the steep 700 nm transition; Narrow NIR B8A provides a nearby 20 m NIR reference.

Arctic value

Useful as a complement to NDVI in greener tundra patches and where subtle pigment or canopy-condition differences matter.



Not universally “better” than NDVI: sparse cover and substrate can still dominate · native 20 m bands

Conceptual spectrum. NDRE should be interpreted as a relative spectral proxy unless calibrated with independent observations.

Beyond NDVI I — greenness and soil-background indices

Choose these when canopy density or exposed substrate changes the behaviour of NDVI

EVI

Enhanced Vegetation Index

$$2.5 \times (\text{NIR B8} - \text{Red B4}) / (\text{NIR B8} + 6 \text{ Red B4} - 7.5 \text{ Blue B2} + 1)$$

[Use it to map / compare](#)

Map relative canopy greenness in productive / high-biomass vegetation where NDVI can lose sensitivity; the Blue B2 term helps reduce some atmospheric/background effects.

EVI2

Two-band Enhanced Vegetation Index

$$2.5 \times (\text{NIR B8} - \text{Red B4}) / (\text{NIR B8} + 2.4 \text{ Red B4} + 1)$$

[Use it to map / compare](#)

Use for EVI-like greenness sensitivity without Blue B2; useful in time series when a simpler two-band formulation is preferred.

SAVI

Soil-Adjusted Vegetation Index

$$1.5 \times (\text{NIR B8} - \text{Red B4}) / (\text{NIR B8} + \text{Red B4} + 0.5)$$

[Use it to map / compare](#)

Map vegetation where exposed soil or mineral substrate contributes strongly to the pixel; often relevant for sparse tundra.

MSAVI2

Modified Soil-Adjusted Vegetation Index 2

$$[2 \text{ NIR B8} + 1 - \sqrt{((2 \text{ NIR B8} + 1)^2 - 8(\text{NIR B8} - \text{Red B4}))}] / 2$$

[Use it to map / compare](#)

Use as a sensitivity check in very sparse or patchy cover; the soil correction adapts to the observed reflectance rather than using a fixed term.

Arctic note: SAVI / MSAVI2 are worth testing when sparse vegetation and exposed substrate dominate the pixel background.

Beyond NDVI II — red edge, moisture, disturbance and snow

Sentinel-2 lets us separate questions that a single greenness index cannot

NDRE

Normalised Difference Red Edge Index

$(\text{Narrow NIR B8A} - \text{Red edge 1 B5}) / (\text{Narrow NIR B8A} + \text{Red edge 1 B5})$

Relative pigment / canopy-condition differences; useful as a complement to NDVI in vegetation mapping.

CIred-edge

Chlorophyll Index Red Edge

$\text{Narrow NIR B8A} / \text{Red edge 1 B5} - 1$

Chlorophyll-sensitive relative signal with a wide dynamic range; interpret comparatively unless calibrated.

NDMI

Normalised Difference Moisture Index

$(\text{Narrow NIR B8A} - \text{SWIR1 B11}) / (\text{Narrow NIR B8A} + \text{SWIR1 B11})$

Moisture-sensitive vegetation / surface context; useful for drying and seasonal water-content differences.

NBR

Normalised Burn Ratio

$(\text{NIR B8} - \text{SWIR2 B12}) / (\text{NIR B8} + \text{SWIR2 B12})$

Strong disturbance and dry-material response; interpret together with greenness, moisture and disturbance history.

NDSI

Normalised Difference Snow Index

$(\text{Green B3} - \text{SWIR1 B11}) / (\text{Green B3} + \text{SWIR1 B11})$

Snow / ice discrimination. Not a vegetation index, but essential to avoid calling snow disappearance "green-up".

Technical detail: any index using Red edge B5 / Narrow NIR B8A / SWIR B11–B12 is based on native 20 m measurements.

Choose the index from the question

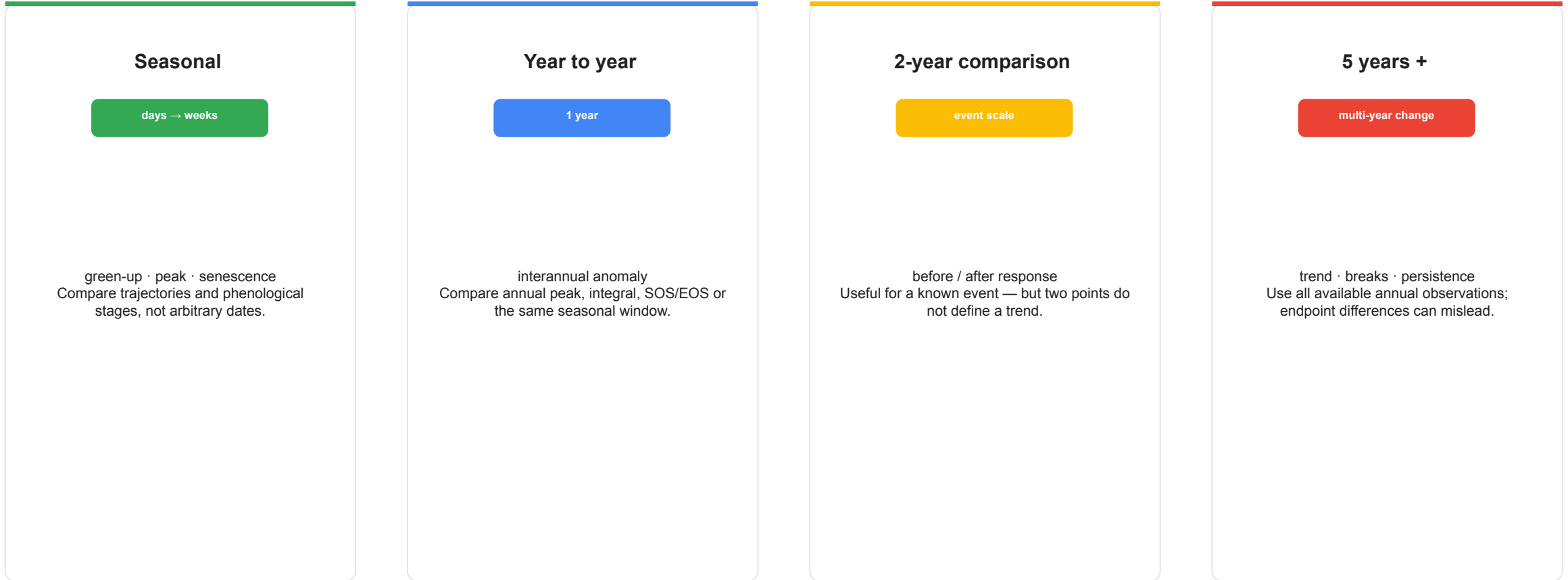
A small, question-driven metric set is usually better than calculating everything

Green vegetation amount seasonal baseline	→	NDVI	Near-infrared B8 + Red B4	Default baseline; inspect snow and substrate.
Dense / productive canopy greenness	→	EVI / EVI2	NIR B8 + Red B4 (+ Blue B2 for EVI)	More dynamic range where NDVI loses sensitivity.
Sparse tundra strong background	→	SAVI / MSAVI2	NIR B8 + Red B4 + soil adjustment	Reduce exposed-substrate influence.
Pigment / canopy condition	→	NDRE / Clred-edge	Red edge B5 + Narrow NIR B8A	Use Sentinel-2 red-edge sensitivity.
Drying / moisture context	→	NDMI	Narrow NIR B8A + SWIR1 B11	SWIR adds water-content sensitivity.
Strong disturbance or dry material	→	NBR + NDVI / NDMI	NIR B8 + SWIR2 B12	Interpret disturbance with spectral context.
Snow versus vegetation signal	→	NDSI + QA	Green B3 + SWIR1 B11	Screen snow before phenology.

Before final selection: inspect matched-date scatter plots + Pearson / Spearman correlations. Keep complementary, not redundant, metrics.

Vegetation change depends on the temporal scale

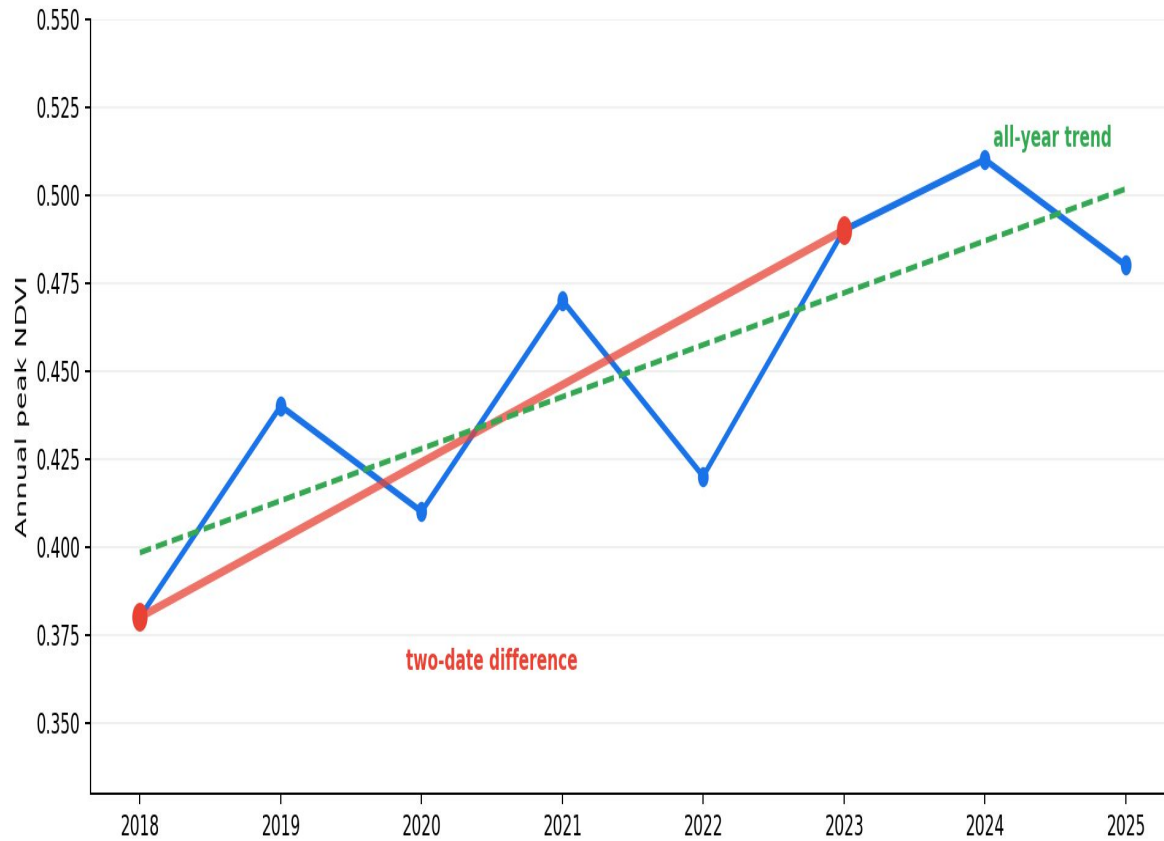
“Change” means something different within a season, between years and across a trend



Critical rule: compare the same season / day-of-year or the same phenological stage. Otherwise seasonality can masquerade as long-term change.

From two-date difference to time-series evidence

Modern change analysis separates seasonality, anomaly, trend and abrupt change



Two images five years apart are snapshots — not a five-year trend.

Δ index map

$\Delta VI = VI(t_2) - VI(t_1)$. Simple and useful for a controlled before/after design.

Seasonal anomaly

Observation – expected value for that time of year. Useful for unusual greening or browning.

Trend

Fit repeated annual summaries or full time-series metrics; report magnitude and uncertainty.

Break / change point

BFAST-type methods separate seasonal behaviour from abrupt or gradual structural change.

Phenology adds “when” to vegetation change

Land-surface phenology describes the seasonal timing of the satellite vegetation signal

Plant phenology

leaf emergence
flowering
leaf senescence



aggregation + mixed pixel

Land-surface phenology

seasonal behaviour of the pixel-level spectral signal

Climate response

Earlier or later seasonal development can reflect temperature and snow conditions.

Growing season

Timing allows us to measure how long vegetation remains seasonally active.

Greening / browning

Phenology helps separate a timing shift from a real change in peak or integrated greenness.

Disturbance / recovery

Anomalous timing or amplitude can reveal stress or recovery — after QA and validation.

Mixed-pixel reminder

A 10 m pixel can contain:

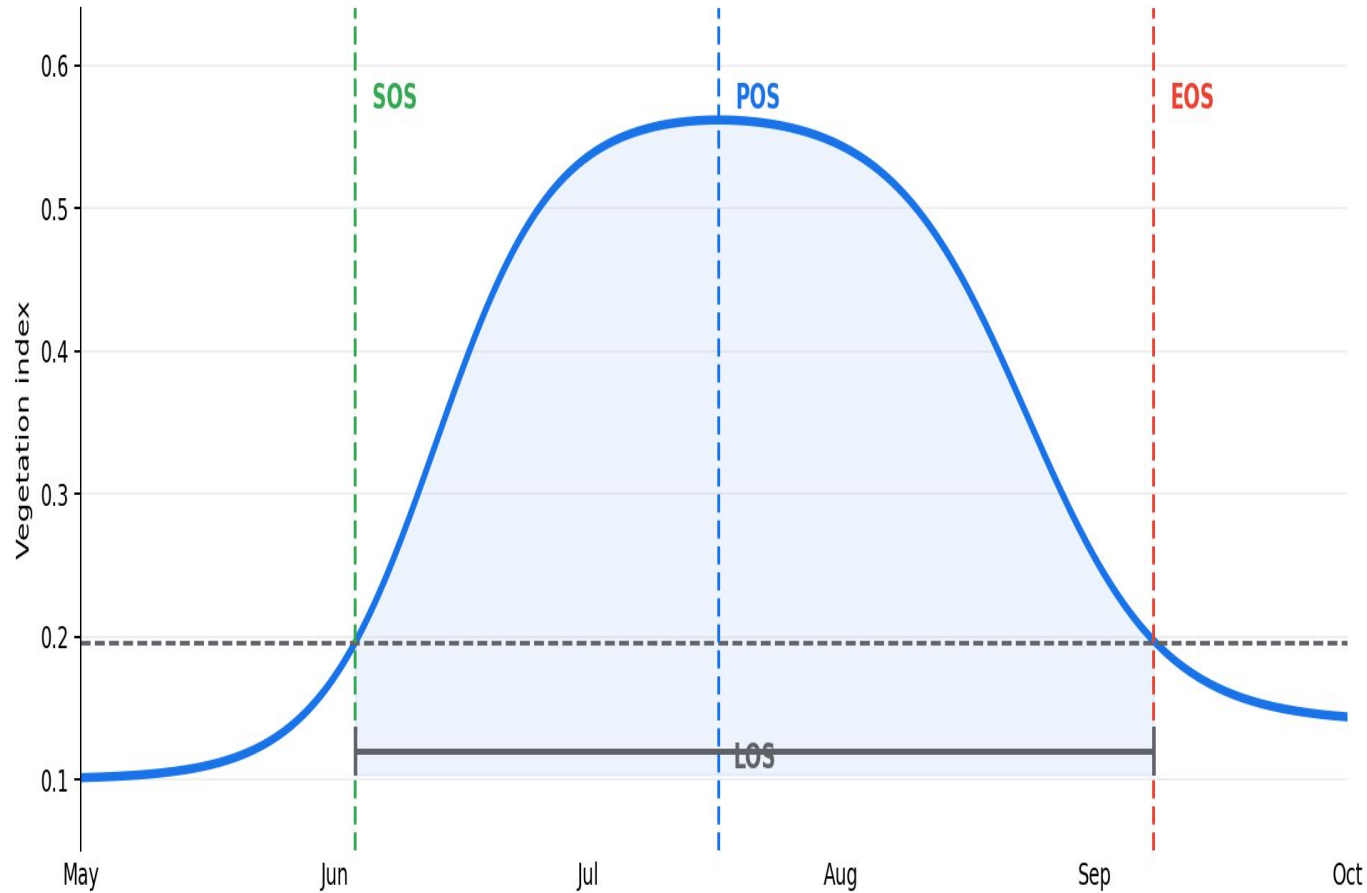
vascular plants
moss
lichen
bare soil
water
snow

Phenology describes the integrated surface signal.

Phenology does not replace change analysis — it explains which part of the seasonal cycle changed.

Phenology metrics

A small set of metrics summarises timing, magnitude and seasonal development



The metric depends on the process we need to explain — not on which output is easiest to calculate.

SOS — Start of Season

When does sustained seasonal activity begin?

POS — Peak of Season

When is maximum seasonal greenness reached?

EOS — End of Season

When does sustained seasonal activity end?

LOS — Length of Season

How long is the active season? $EOS - SOS$

Peak + amplitude

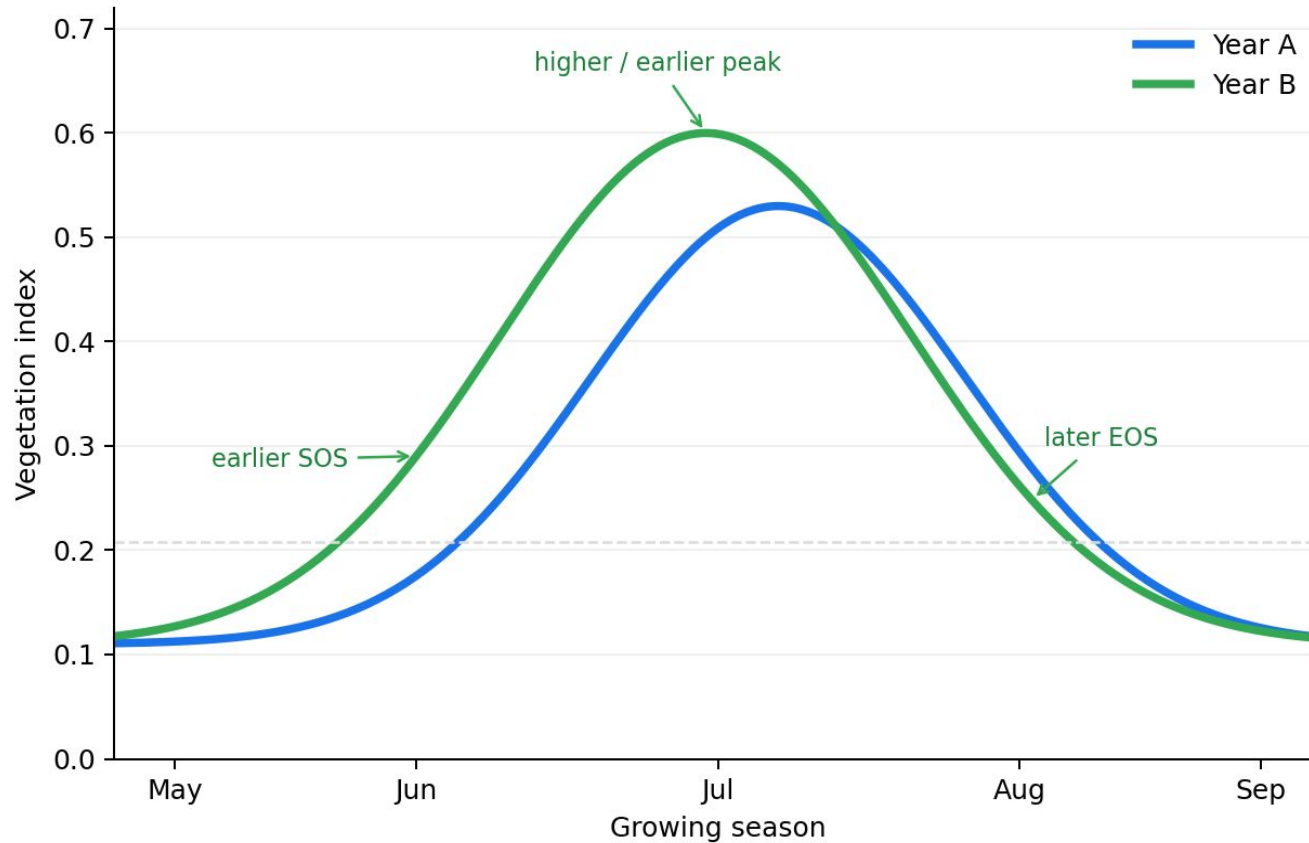
How strong is the maximum and seasonal contrast?

Rates + seasonal integral

How quickly does the signal change, and how much accumulates through the season?

What phenological change can tell us

The same “greening” signal can arise from different changes in timing and magnitude



Earlier SOS

The seasonal rise begins sooner — often linked to snow and spring conditions.

Longer LOS / later EOS

The seasonal signal persists for longer or ends later.

Higher peak or amplitude

A stronger maximum greenness signal — not automatically more biomass.

Lower peak / browning

Could reflect stress, herbivory, disturbance, composition change or poor peak sampling.

Phenology helps separate a change in timing from a change in seasonal magnitude.

Arctic time series: quality control is part of the science

In Svalbard, snow and observation conditions can be as important as the vegetation signal

Cloud / fog

Missing observations and residual contamination.

Snow / ice

A strong seasonal transition that can mimic green-up.

Low solar angle

Illumination and BRDF effects increase late / early in season.

Topographic shadow

Locally suppresses reflectance and index values.

Short growing season

Losing a few dates can shift SOS / EOS substantially.

Mixed tundra pixels

Vegetation, soil, moss, water and snow contribute together.

L2A



Cloud Score+



snow screen



indices



time series

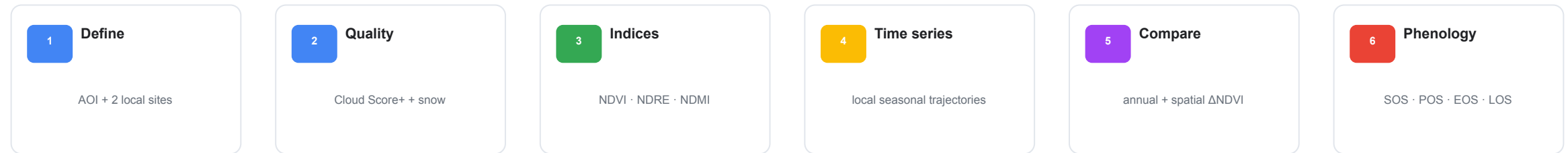


metrics + uncertainty

A smooth curve can look convincing even when the underlying observations are sparse.

Practical: one Sentinel-2 workflow, several vegetation questions

The Colab exercise uses the same Adventdalen AOI as Block 1 and local sites for time series



The goal is not to calculate the largest number of indices. The goal is to explain why the selected metric answers the scientific question.